Scalability and Energy Structural Constraints in Artificial Intelligence

Revisiting Sutton's Bitter Lesson in Large-Scale AI Systems

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Abstract

This paper revisits Richard Sutton's Bitter Lesson through the Energetic Paradigm, arguing that scalability alone is insufficient for future AI systems. As models reach extreme scale, energy consumption, load concentration, and infrastructural constraints become decisive factors. We formalize capability growth under energy constraints and show that sustainable AI development requires optimizing not only performance but also energetic efficiency and structural stability.

A. A Rarely Recognized Trajectory in Artificial Intelligence

Within mainstream historical narratives of artificial intelligence, Richard Sutton occupies an unusual position. He was neither among the earliest researchers developing neural networks, nor the builder of the largest models, nor the initiator of large-scale datasets.

However, if one carefully traces the historical trajectory of artificial intelligence, a remarkably stable intellectual line becomes visible. Across multiple technological generations, the central idea of this trajectory can be traced to the work of Richard Sutton.

The purpose of this paper is not merely to introduce Sutton's work. Instead, it asks a more direct question: why have Sutton's ideas become even clearer after more than a decade of rapid progress in deep learning? Furthermore, what emerges if Sutton's ideas are interpreted through the framework of the Energetic Paradigm?

B. From Rule-Based Systems to Scalable Methods

During the 1980s, the dominant paradigm in artificial intelligence was knowledge engineering. Systems developed during this period were commonly referred to as expert systems.

The core assumption underlying expert systems was straightforward: if rules were specified in sufficient detail, the system could perform reasoning tasks effectively. Formally, such systems attempted to encode domain knowledge explicitly through rule sets:

$$Decision = f(R_1, R_2, \dots, R_n)$$

where R_i represents handcrafted rules derived from expert knowledge.

However, this approach relied on a strong implicit assumption: that human experts were capable of explicitly articulating the knowledge required for intelligent reasoning. Sutton recognized early on that this assumption was fragile.

His critique was not directed at rules themselves. Rather, it concerned the idea that the growth of machine intelligence should depend on the speed at which humans are able to organize and encode knowledge.

This insight eventually became known as **The Bitter Lesson**.

The argument of the Bitter Lesson is relatively simple. Over long time horizons, the most effective AI methods are those that leverage large-scale computation and scale naturally with increasing computational resources.

In other words, instead of embedding human knowledge directly into models, general learning methods allow models to improve automatically as the scale of data and computation increases. Such methods do not rely on domain-specific knowledge but instead approach optimal solutions through general optimization mechanisms.

This position is remarkably restrained. It does not rely on clever manual design; rather, it places its confidence in scalability.

From this perspective, the breakthrough of ImageNet in 2012 was not accidental. On the foundation of the ImageNet dataset created by Fei-Fei Li, the AlexNet model developed by Geoffrey Hinton's group dramatically increased both model scale and computational usage. This shift brought deep learning into widespread prominence.

In retrospect, Sutton had effectively anticipated this development: scalable computational methods would ultimately dominate rule-based approaches.

C. Reinforcement Learning as a General Method

Sutton's primary research domain is reinforcement learning. Together with Andrew Barto, he established the theoretical foundations of modern reinforcement learning.

Unlike supervised learning, which focuses primarily on classification or generation tasks, reinforcement learning addresses the problem of decision-making. In a dynamic environment, an agent observes the environment, selects actions, and receives feedback from the consequences of those actions.

Formally, the objective of reinforcement learning is to maximize the expected cumulative return

$$R_t = \sum_{k=0}^{\infty} \gamma^k \, r_{t+k}$$

where r_t denotes the reward at time $t$and γ is a discount factor.

Learning occurs through iterative updates of value functions and policies. For example, a typical temporal-difference update can be expressed as

$$V(s_t) \leftarrow V(s_t) + \alpha(r_t + \gamma V(s_{t+1}) - V(s_t))$$

Through repeated interaction with the environment, the agent gradually adjusts its policy to approach an optimal strategy.

This framework tightly couples learning with action. It therefore represents a dynamic perspective on intelligence.

D. Long-Horizon Decision Frameworks: AlphaGo and RLHF

A defining feature of reinforcement learning is that it evaluates actions according to their long-term consequences. This long-horizon perspective later became central to several influential AI systems.

One prominent example is AlphaGo and its successor AlphaZero. These systems did not rely on memorized human strategies. Instead, they improved by repeatedly playing games against themselves and updating their strategies based on win-loss feedback.

A similar principle also appears in modern large language models through **Reinforcement Learning from Human Feedback (RLHF)**. In RLHF systems, models first generate candidate responses. Human evaluators then assign preference scores to these responses. Reinforcement learning algorithms subsequently adjust the model's output policy in order to align future responses more closely with human preferences.

Sutton's contribution therefore does not lie in any single model architecture. Rather, he consistently emphasized a broader principle: intelligence should be understood as a process of continuously optimizing long-term return under uncertainty.

This represents a fundamentally dynamic view of learning systems.

E. A Constraint Sutton Did Not Need to Confront

Why do Sutton's ideas appear even clearer today? The answer lies in the scale of contemporary AI systems.

Modern models operate at unprecedented scales. Large language models with hundreds of billions or even trillions of parameters are capable of generating video, writing software, and simulating complex conversations.

Yet the underlying driver of these capabilities remains the same: scale. Increasing numbers of parameters, larger datasets, and greater computational resources all contribute to improved performance.

This pattern aligns closely with Sutton's Bitter Lesson. Methods that scale with computational resources tend to dominate in the long run. As models grow larger and rely more heavily on data and computation, the advantages of general methods become increasingly evident. Approaches based on manually engineered rules struggle to maintain their effectiveness as scale increases.

However, contemporary AI development introduces a constraint that Sutton did not need to confront directly.

Sutton's notion of scalability implicitly assumed that increasing computational resources would always translate into improved performance, and that computational resources themselves would not become structural bottlenecks.

Today, this assumption no longer holds.

Even when algorithms remain scalable, the acquisition, distribution, and scheduling of computation may become limiting factors. The issue is no longer simply whether algorithms can utilize additional computational resources. Instead, the question becomes whether the system can sustain the continued expansion of computation itself.

When model scale approaches its practical limits, the central issue shifts. The question is no longer merely whether further expansion is possible, but whether such expansion remains feasible within the physical and structural constraints of the system.

This challenge is already visible in contemporary AI infrastructure. When models reach hundreds of billions or trillions of parameters, training costs rise dramatically. Inference costs are no longer determined primarily by model size but increasingly by memory bandwidth and data transfer latency. Moreover, the energy consumption and thermal constraints of data centers become significant operational limitations.

Thus, increasing scale is no longer purely a performance advantage. It simultaneously reshapes the cost structure of artificial intelligence systems.

F. Reading Sutton Through the Energetic Paradigm

At this point, Sutton's argument can be reformulated in a more formal way. The Bitter Lesson can be interpreted as a statement about resource allocation: instead of allocating limited design effort to handcrafted rules, one allocates computational resources to scalable learning mechanisms. In ordinary discussion, this is often described qualitatively. Under the Energetic Paradigm, however, the same claim can be written as an allocation problem over computational energy, system load, and long-horizon return.

Let the total system energy budget be denoted by

$$E_{\text{tot}} = E_{\text{train}} + E_{\text{infer}} + E_{\text{move}} + E_{\text{cool}} + E_{\text{sync}},$$

where E_{train} is the energy spent on parameter updates, E_{infer} is inference energy, E_{move} is the energy cost of data movement, E_{cool} is thermal management cost, and E_{sync} is the cost of global coordination across distributed nodes. In small systems, the dominant term is often training compute itself. In large-scale systems, however, non-compute terms cease to be negligible and often become structurally decisive.

Sutton's scalability principle may then be written as a preference for methods whose capability C increases monotonically with computational investment K:

$$\frac{\partial C}{\partial K} > 0.$$

This captures the core intuition of scalable methods, but it remains incomplete because it does not specify the energetic structure required to sustain that increase. Under the Energetic Paradigm, the relevant quantity is not only capability growth, but capability growth per unit energetic burden. A more informative object is therefore

$$\eta_C = \frac{\Delta C}{\Delta E_{\text{tot}}},$$

which measures capability gain per marginal increase in total system energy. Once this ratio declines sharply, additional scale may still improve performance, but it no longer does so in an energetically healthy way.

Sutton's position can therefore be reinterpreted as follows: replace human rule-construction cost with scalable optimization cost. Let Rdenote handcrafted rule investment and Ldenote learning-system investment. Then the classical rule-based paradigm implicitly seeks capability through

$$C \approx f(R),$$

whereas the scalable paradigm seeks capability through

$$C \approx g(D, K, \Theta),$$

where D is data scale, K is computational scale, and Θdenotes trainable parameters. The Energetic Paradigm adds a missing layer by requiring that the induced energy burden remain structurally admissible:

$$C \approx g(D, K, \Theta)\text{subject to}E_{\text{tot}} \leq E_{\max}, L_{\text{sys}} \leq L_{\max}, S_{\text{stab}} \geq S_{\min}.$$

Here L_{sys} is system load and S_{stab} is a stability functional. In this form, scalability is no longer an unconstrained growth principle. It becomes a constrained optimization problem over capability, energy, and stability simultaneously.

Reinforcement learning already provides part of this formal machinery. In standard form, the objective is to maximize expected discounted return:

$$J(\pi) = \mathbb{E}_{\pi}[\sum_{t=0}^{\infty} \gamma^t r_t].$$

This objective assumes that the relevant criterion is cumulative reward. The Energetic Paradigm modifies the problem by introducing energetic dissipation and structural load into the same target. Let $e_t$denote instantaneous energy expenditure, $m_t$data-movement cost, and $\ell_t$system load penalty. Then the optimization target becomes

$$J_{\text{EP}}(\pi) = \mathbb{E}_{\pi}[\sum_{t=0}^{\infty} \gamma^t (r_t - \alpha e_t - \beta m_t - \delta \ell_t)].$$

This is the first place where the mathematics matters. Once energy cost enters the same objective as reward, the gradient field changes. The system is no longer driven only toward high-return trajectories, but toward trajectories with better return-to-energy structure.

Under policy-gradient learning, the standard update direction is

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\, \hat{A}_t].$$

After the Energetic Paradigm correction, the effective advantage term is no longer merely reward-based. It becomes

$$\hat{A}_t^{\mathrm{EP}} = \hat{A}_t - \alpha \hat{E}_t - \beta \widehat{M}_t - \delta \hat{L}_t,$$

so that

$$\nabla_\theta J_{\mathrm{EP}}(\pi_\theta) = \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\, \hat{A}_t^{\mathrm{EP}}].$$

This means that a trajectory with high reward but extreme energy burden no longer receives the same positive update pressure. A high-return path can be suppressed if its energetic dissipation and system stress are too large. Conversely, a path with slightly lower reward but much lower energetic burden can become dominant in training.

This is the formal meaning of the earlier qualitative claim that the Energetic Paradigm does not merely ask whether more resources should be allocated to learning, but whether the resulting allocation remains energy-efficient under scale amplification. The question is no longer

$$\max_{\pi} \quad \mathbb{E}[R],$$

but

$$\max_{\pi} \quad \mathbb{E}[R] \text{subject to} \mathbb{E}[E_{\mathrm{tot}}] \le \bar{E}, \mathbb{E}[L_{\mathrm{sys}}] \le \bar{L}, \Pr\,(\text{instability}) \le \varepsilon.$$

In other words, the optimization target is lifted from mere scalability to admissible scalability.

G. From the Bitter Lesson to Load Structure

Once the problem is written in this way, the next step is to show why large-scale systems force a transition from performance analysis to load-structure analysis. The

Bitter Lesson says that long-run success belongs to methods that exploit computation. That statement remains correct, but only under the hidden assumption that computation can be expanded without destabilizing the system that carries it.

Let total capability be $C(N)$, where N denotes system scale. Sutton's thesis focuses on conditions under which

$$\frac{dC}{dN} > 0.$$

The missing question is whether the same scaling process also drives load concentration and structural stress upward:

$$\frac{dL_{\text{sys}}}{dN} > 0, \frac{dE_{\text{tot}}}{dN} > 0, \frac{dH_{\text{conc}}}{dN} > 0.$$

Here H_{conc} denotes a concentration measure over load or energy density across system nodes. For example, if the distributed infrastructure contains nodes $i = 1, \dots, n$, with node-level energy burdens p_i, then concentration can be measured as

$$H_{\text{conc}} = \sum_{i=1}^{n} p_i^2, p_i = \frac{E_i}{\sum_{j=1}^{n} E_j}.$$

When H_{conc} rises, the system is no longer merely scaling; it is concentrating energetic pressure into fewer locations. That concentration produces bottlenecks in memory bandwidth, synchronization, thermal dissipation, and data transport. At that point, the relevant object is not raw performance growth but the topology of the load-bearing structure.

This leads to a more precise formulation of healthy scaling. Scale expansion is not simply an increase in N. It is an increase in N that preserves bounded structural stress:

$$\limsup_{N\to\infty} L_{\text{sys}}(N) < \infty, \limsup_{N\to\infty} H_{\text{conc}}(N) < \infty, \eta_C(N) = \frac{dC/dN}{dE_{\text{tot}}/dN} \to /0.$$

If these conditions fail, one still obtains a larger model, but not a structurally sustainable one.

The real dividing line, then, is not whether models should continue to grow. It is whether growth preserves stability. Let $S(N)$ be a structural stability functional. Then the relevant criterion is not simply

$$C(N+1) > C(N),$$

but rather

$$C(N+1) > C(N) \text{ and } S(N+1) \geq S_{\min}.$$

If scale growth increases capability while driving $S(N)$ toward a critical threshold, the system is moving toward imbalance. In such a regime, the central issue is no longer performance improvement, but proximity to instability:

$$S(N) \downarrow S_{\text{crit}}.$$

The problem can therefore be restated in formal terms: how can scale continue increasing while preserving a sufficient margin from critical instability? That is,

$$\max_N \; C(N) \text{subject to} S(N) - S_{\text{crit}} \geq \Delta_{\text{safe}}.$$

This is the mathematical form of the claim that technological expansion must preserve a civilizational throttle and a system-level capacity for self-regulation.

H. What the Energetic Paradigm Does

Sutton solves the problem of how learning mechanisms improve as computational scale grows. The Energetic Paradigm studies a different layer: whether the energy flows induced by that scaling process remain stable, whether cost, load, and risk rise synchronously, and whether capability improvement is purchased at the price of structural imbalance.

This means that the object of analysis is the energy structure of expansion itself. Let the system be represented by a directed graph $G = (V, \mathcal{E})$, where nodes are computational or infrastructural units and edges are channels of transfer. Let f_{ij}denote energy flow from node $i$to node j. Then the system's energetic structure is characterized by

$$\sum_j f_{ij} - \sum_k f_{ki} = b_i,$$

where b_i is net injection or extraction at node i. Stability requires that these flows remain within carrying capacity constraints:

$$f_{ij} \leq \kappa_{ij}, \forall (i,j) \in \mathcal{E}.$$

The Energetic Paradigm asks three formal questions.

First, where does the operative energy come from? This is a source problem, represented by the injection vector b.

Second, how does energy propagate? This is a transport problem, represented by the flow field f_{ij}.

Third, does energy become excessively concentrated? This is a concentration-risk problem, represented by quantities such as H_{conc}, maximal node burden $\max_i E_i$, or spectral stress measures over G.

I. How the Energetic Paradigm Operates

The Energetic Paradigm is not an abstract criticism. It converts the problem into analyzable variables. Let

$$\rho_E = \frac{E_{\text{tot}}}{C}$$

denote energy density, that is, the energetic cost per unit capability. Let

$$\Lambda = \max_i \; p_i$$

denote maximal load concentration. Let S_{stab} denote the recoverable stability margin, and let D_{path} denote path dependence, which can be approximated by hysteresis or irreversibility measures in the dynamics.

The core analytical question is then: how do these variables change with scale N?

If capability scales approximately linearly,

$$C(N) \sim aN,$$

but total energy cost scales superlinearly,

$$E_{\text{tot}}(N) \sim bN^{\kappa}, \kappa > 1,$$

then

$$\rho_E(N) = \frac{E_{\text{tot}}(N)}{C(N)} \sim \frac{b}{a} N^{\kappa-1},$$

which grows with N. In that case, expansion is not healthy expansion. Capability is increasing, but energetic cost per unit capability is worsening.

The role of the Energetic Paradigm is therefore not to replace reinforcement learning, but to add an energy-constraint layer to it. Formally, this means changing the feasible region of learning dynamics from an unconstrained optimization problem to a constrained one:

$$\max_{\pi} \quad J(\pi) \rightsquigarrow \max_{\pi} \quad J_{\text{EP}}(\pi)$$

subject to energetic and structural admissibility constraints.

J. Why the Energetic Paradigm Can Do This

The reason the Energetic Paradigm can accomplish this is simple in formal terms. Any optimizable system follows its objective gradient. If the objective contains only reward, then the dynamics push behavior only toward maximal return. If energy cost is inserted into the same objective, the gradient field necessarily changes.

Let the original objective be

$$\mathcal{L}_0(\theta) = -J(\theta).$$

After energetic correction, define

$$\mathcal{L}_{\text{EP}}(\theta) = -J(\theta) + \alpha E(\theta) + \beta L(\theta) + \delta R_{\text{risk}}(\theta),$$

where $E(\theta)$ is total energetic burden, $L(\theta)$ is system load, and $R_{\text{risk}}(\theta)$ is structural risk. Then

$$\nabla_\theta \mathcal{L}_{\text{EP}} = -\nabla_\theta J + \alpha \nabla_\theta E + \beta \nabla_\theta L + \delta \nabla_\theta R_{\text{risk}}.$$

This is not rhetoric. It means that high-reward but highly dissipative paths lose gradient attractiveness, while lower-cost paths gain it. Once that occurs, the system no longer expands blindly. It optimizes along energetically cheaper and structurally more stable trajectories.

That is the actual mathematical content of the claim that the Energetic Paradigm turns "the most energy-efficient path" into an endogenous direction of learning and decision-making.

K. Interpreting the Energetic Paradigm Through Complex Systems

The Energetic Paradigm can also be interpreted through the lens of **complex systems theory**. Within this theoretical framework, what is commonly referred to as an "optimal solution" is rarely the result of a centralized search process. Instead, it emerges as a **stable structure selected over time under constraints, energy flows, and cumulative dynamics**.

Formally, a complex system can be represented as a collection of interacting components. Let the system consist of Ninteracting units x_i, whose dynamics evolve according to

$$x_i(t+1) = F_i(x_i(t), \mathcal{N}_i(t), \xi_i(t)),$$

where $\mathcal{N}_i$ denotes the local neighborhood of interactions and ξ_i represents stochastic perturbations or environmental inputs. No global controller explicitly searches for an optimal configuration. Instead, system-level structure emerges from the interactions among local rules.

Each local action generates measurable costs. These costs may take the form of energy dissipation, resource expenditure, temporal delay, or accumulated systemic risk. Let the instantaneous cost of a system state be

$$C(t) = E(t) + R(t) + D(t) + \Gamma(t),$$

where $E(t)$ represents energy dissipation, $R(t)$ resource consumption, $D(t)$ delay or latency, and $\Gamma(t)$ risk accumulation. When these costs are real system variables rather than ignored externalities, the dynamics of the system begin to exhibit **selection effects**.

Over repeated operation, trajectories with high dissipation or structural tension lose viability. Conversely, trajectories that maintain lower dissipation and lower systemic stress gradually accumulate advantage. In dynamical systems language, the system begins to converge toward **stable attractors**.

Let the state vector of the system be $X(t)$. The long-term evolution of the system can then be described as convergence toward an attractor set $\mathcal{A}$:

$$\lim_{t \to \infty} X(t) \in \mathcal{A}.$$

These attractors correspond to configurations that remain stable under the system's energy throughput and structural constraints.

A crucial component in this process is **time**. A single successful action does not guarantee long-term persistence. However, when the system operates continuously, any structure that cannot sustain the ongoing flow of energy will eventually collapse or be replaced.

Let $S(X)$ denote the structural stability of configuration X. A configuration remains viable only if

$$S(X) \geq S_{\text{crit}},$$

where S_{crit} represents the critical threshold below which instability occurs. Configurations that fall below this threshold disappear through the cumulative effect of resource and energy constraints.

The apparent movement of complex systems toward "optimal" configurations therefore has a precise interpretation. It is not the result of foresight or centralized optimization. Instead, it arises because only structures close to **minimum dissipation or maximal stability** can persist over long time horizons.

L. Learning Systems as Energy-Filtered Dynamics

When this principle is applied to learning systems, its internal consistency becomes clear. Consider a reinforcement learning system whose objective is to maximize expected cumulative reward:

$$J(\pi) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right].$$

If the optimization process considers only reward and ignores energy expenditure, the learning dynamics may favor strategies with high short-term reward but extremely high energy costs.

Let the energy expenditure of an action at time t be e_t. If the energy term is not included in the objective, the optimization process may prefer trajectories satisfying

$$r_t \gg 0 \text{even when} e_t \gg 0.$$

Such strategies may appear optimal under purely reward-based optimization but may fail when confronted with real-world resource constraints.

The Energetic Paradigm modifies this objective by introducing energetic dissipation directly into the optimization target:

$$J_{\mathrm{EP}}(\pi) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^{t}\,(r_t - \alpha e_t)\right].$$

Once the energetic penalty term αe_t is included, the gradient dynamics change. High-energy trajectories are penalized during policy updates, while lower-dissipation trajectories gradually accumulate advantage.

Let the policy parameters be θ. The resulting update direction becomes

$$\nabla_{\theta} J_{\mathrm{EP}} = \mathbb{E}[\nabla_{\theta} \log\, \pi_{\theta}(a_t \mid s_t)(A_t - \alpha E_t)].$$

Under this update rule, the system does not converge toward the mathematically perfect reward-maximizing strategy. Instead, it converges toward a strategy that is **stable and sustainable under the system's energy throughput constraints**.

Formally, the resulting equilibrium state $\pi^{\backslash *}$satisfies

$$\pi^{\backslash *} = \arg\, \max_{\pi}\; \mathbb{E}_{\pi}[R] \text{subject to} \mathbb{E}_{\pi}[E] \leq E_{\max}.$$

The resulting equilibrium is therefore not a purely mathematical optimum but a **steady-state configuration** compatible with the system's energy flux and structural limits.

M. Stability as the Fundamental Selection Mechanism

From this perspective, the apparent "self-optimization" of complex systems can be understood as the outcome of a stability-based filtering mechanism.

Let the system possess an energy flux Φ. Changes in energy throughput alter the set of stable configurations:

$$\mathcal{A}(\Phi) = \{X \mid S(X, \Phi) \geq S_{\mathrm{crit}}\}.$$

Increasing flux may destabilize previously stable configurations by increasing dissipation or concentrating system load. Likewise, excessive concentration of load within particular nodes introduces structural risk.

If the system consists of nodes $i = 1, \dots, N$ with node-level loads L_i, a concentration index may be defined as

$$H = \sum_{i=1}^{N} \left(\frac{L_i}{\sum_{j=1}^{N} L_j} \right)^2 .$$

Higher values of H indicate stronger load concentration. As concentration increases, the probability of structural failure rises because a small subset of nodes bears disproportionate system stress.

Within the Energetic Paradigm, these variables—energy flux, dissipation, and load concentration—constitute the physical basis of the selection mechanism governing system evolution.

When these variables are made explicit and incorporated into system design, the trajectory of system evolution is no longer characterized by blind expansion. Instead, it follows the subset of trajectories that remain **energetically admissible and structurally stable**.

Thus, complex systems appear to "optimize themselves" not because they possess intentional goals, but because unstable configurations cannot persist. Stability functions as the most powerful selection mechanism, and **energy flow forms the underlying substrate through which that mechanism operates**.

N. Conclusion

This paper revisited Richard Sutton's **Bitter Lesson** through the lens of the **Energetic Paradigm**, with the goal of clarifying both the enduring validity of Sutton's argument and the structural limitations that emerge as artificial intelligence systems reach unprecedented scale.

Sutton's central claim—that long-term progress in artificial intelligence is driven by **general methods capable of exploiting large-scale computation**—has been repeatedly validated by empirical developments. From the ImageNet breakthrough to the emergence of large language models, the dominant trajectory of AI research has consistently favored methods that improve as data and computational resources increase. In this sense, Sutton's insight remains fundamentally correct.

However, the expansion of contemporary AI systems introduces constraints that were not central in the earlier stages of AI development. As models approach hundreds of billions or trillions of parameters, computational scaling is no longer merely an algorithmic matter. It becomes an infrastructural phenomenon involving **energy**

consumption, data movement, synchronization costs, and systemic load distribution. Under such conditions, the central question shifts from whether learning methods can scale to whether the **systems carrying that scale remain structurally sustainable**.

The Energetic Paradigm provides a framework for analyzing this transition. Rather than focusing exclusively on reward maximization or predictive accuracy, the framework introduces **energy flow, dissipation, and structural load** as explicit variables in the analysis of learning systems. When these variables are incorporated into optimization objectives, the learning dynamics themselves change. Strategies that maximize reward at extreme energetic cost lose gradient advantage, while strategies that achieve comparable outcomes with lower dissipation gradually become dominant.

From a complex systems perspective, this transformation reveals that apparent optimization in large-scale systems is not the result of centralized design but of **stability-based selection**. Systems operating under continuous energy flux inevitably filter out unstable configurations over time. Only those structures capable of sustaining energy flow within physical and infrastructural constraints remain viable. In this sense, the "optimal solutions" observed in learning systems are better understood as **steady-state attractors under energetic constraints**.

The implication is that scalability alone is not a sufficient design principle for the next generation of artificial intelligence systems. While Sutton's Bitter Lesson correctly emphasizes the importance of general methods and computational expansion, the long-term viability of large-scale AI will increasingly depend on the **energetic structure of that expansion**. Capability growth must therefore be evaluated not only in terms of performance improvements but also in terms of **energy efficiency, load distribution, and structural stability**.

The Energetic Paradigm does not replace reinforcement learning or scalable learning methods. Instead, it introduces an additional analytical layer that constrains and guides their evolution. By embedding energy costs and structural constraints directly into optimization objectives, learning systems can move from unconstrained expansion toward **admissible scalability**—a regime in which capability growth remains compatible with the physical and infrastructural systems that sustain it.

Future research will need to further formalize these energetic constraints and investigate how they interact with distributed computation, large-scale optimization, and long-horizon decision systems. As artificial intelligence continues to expand its role in critical infrastructure and global decision-making processes, understanding the

energetic structure of learning systems may become as essential as understanding their algorithms.